1 Explain the four values of the Agile Manifesto with suitable examples.

2 What are Agile Methods? Explain the classification of Agile methods.

3 Define Agile Project Management. Explain its major characteristics and benefits.

4 Compare the Traditional Model and Agile Model with respect to requirements, planning, development, and customer involvement.

5 Explain Agile Team Interactions and discuss their importance in successful Agile projects.

6 Explain the importance of Ethics in Agile Teams with suitable examples.

1. Four Values of the Agile Manifesto

The **Agile Manifesto** gives four important values that guide Agile software development.

1. Individuals and interactions over processes and tools

Agile gives more importance to **people, communication, and teamwork** than to strict processes or tools.

Example:
If a developer faces a problem, they can directly discuss it with the team instead of waiting for a formal process.

2. Working software over comprehensive documentation

Agile focuses more on delivering **working software** than writing large amounts of documentation.

Example:
Instead of spending two months preparing documents, the team delivers a working part of the application in a few weeks.

3. Customer collaboration over contract negotiation

Agile encourages **continuous communication with the customer** rather than depending only on the initial contract.

Example:
If the customer wants a new feature during development, the team discusses it and adds it to a future sprint.

4. Responding to change over following a plan

Agile accepts that requirements can **change during development**.

Example:
If users request a new payment option, the development team can change the next sprint's work to include it.

Easy way to remember:

People → Software → Customer → Change

Conclusion: The four values help teams develop software quickly, communicate effectively, satisfy customers, and adapt to changes.

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2. What are Agile Methods? Explain the Classification of Agile Methods.

Definition

Agile methods are software development approaches that focus on **iterative development, teamwork, customer feedback, and responding to changing requirements**.

Instead of developing the entire software at once, Agile develops it in **small increments**.

Classification of Agile Methods

Agile methods can be broadly classified into the following:

1. Scrum

Scrum divides development into short periods called **Sprints**, usually 1–4 weeks.

Example:
A team develops login functionality in Sprint 1 and payment functionality in Sprint 2.

2. Extreme Programming (XP)

XP focuses on **high-quality code, continuous testing, pair programming, and frequent releases**.

Example:
Two programmers work together on the same code to reduce errors.

3. Kanban

Kanban uses a **visual board** to track work.

Typical stages are:

To Do → In Progress → Testing → Done

Example:
A team moves a task from "To Do" to "Done" as it progresses.

4. Crystal

Crystal focuses mainly on **communication, teamwork, and people**. Different Crystal methods are used depending on team size and project complexity.

Easy way to remember:

S-X-K-C

- **S** → Scrum
- **X** → XP
- **K** → Kanban
- **C** → Crystal

Conclusion: Agile methods help teams deliver software faster, improve quality, and respond easily to customer changes.

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3. Define Agile Project Management. Explain its Major Characteristics and Benefits.

Definition

Agile Project Management (APM) is a project management approach that uses Agile principles to develop software through **small iterations, continuous feedback, teamwork, and flexibility**.

Major Characteristics

1. Iterative development

The project is divided into **small iterations or sprints**.

2. Customer involvement

Customers regularly provide **feedback** during development.

3. Flexible requirements

Requirements can be **changed when necessary**.

4. Continuous delivery

The team regularly delivers **working software**.

5. Team collaboration

Developers, testers, customers, and managers work **closely together**.

Benefits

- **Faster delivery** of software

- **Better software quality**
- Easy to **handle changing requirements**
- Regular **customer feedback**
- Better **team communication**
- Problems can be identified **early**

Easy way to remember characteristics:

I-C-F-C-T

I → Iterative
C → Customer involvement
F → Flexible
C → Continuous delivery
T → Teamwork

Conclusion: Agile Project Management makes projects more flexible, collaborative, and customer-focused.

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4. Compare Traditional Model and Agile Model

Basis	**Traditional Model**	**Agile Model**
Requirements	Requirements are mostly fixed at the beginning.	Requirements can change during development.
Planning	Detailed planning is done at the beginning.	Planning is done continuously for each iteration.
Development	Software is developed in sequential phases.	Software is developed in small iterations/sprints.
Customer involvement	Customer involvement is mainly at the beginning and end.	Customer is involved continuously.
Changes	Changes are difficult and costly.	Changes are easily accepted.
Delivery	Product is usually delivered at the end.	Working software is delivered frequently.

Simple example

Traditional:
Plan everything → Develop → Test → Deliver

Agile:
Plan → Develop → Test → Get feedback → Improve → Repeat

Easy way to remember:

Traditional = Fixed + Sequential + Less customer involvement

Agile = Flexible + Iterative + Continuous customer involvement

Conclusion: Traditional development works well when requirements are stable, while Agile is better when requirements may change frequently.

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5. Explain Agile Team Interactions and Their Importance

Definition

Agile team interaction means the **regular communication, collaboration, and cooperation** among team members, customers, and stakeholders during Agile development.

Agile teams work together rather than working separately.

Important Team Interactions

1. Daily Stand-up

Team members have a short daily meeting to discuss:

- What they completed
- What they will do next
- Any problems they are facing

2. Collaboration

Developers, testers, designers, and other members work together to complete tasks.

3. Customer interaction

The team regularly communicates with the customer to understand requirements and receive feedback.

4. Sprint Review

At the end of a sprint, the team demonstrates the completed work to the customer and stakeholders.

5. Sprint Retrospective

The team discusses **what went well, what went wrong, and how to improve**.

Importance

- Improves **communication**
- Builds better **teamwork**
- Helps identify problems **early**
- Provides quick **feedback**
- Improves **software quality**
- Helps complete projects successfully

Easy way to remember:

Daily → Collaborate → Customer → Review → Improve

Conclusion: Good team interaction is important because Agile depends heavily on communication, teamwork, feedback, and continuous improvement.

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6. Importance of Ethics in Agile Teams

Definition

Ethics in Agile teams means following **honest, responsible, fair, and professional behavior** while working with team members and customers.

Ethical behavior creates trust and helps the team work effectively.

Importance of Ethics

1. Honesty

Team members should honestly report the actual progress of the project.

Example:
If a task is incomplete, the developer should say so instead of saying it is completed.

2. Respect

Every team member should respect the **ideas and opinions of others**.

Example:
A developer should listen to a tester's concerns instead of ignoring them.

3. Transparency

Project information should be shared openly with the team and customer.

Example:
If there is a major delay, the customer should be informed.

4. Responsibility

Team members should take responsibility for their assigned work.

Example:
A developer should fix a bug they introduced instead of blaming another team member.

5. Customer interest

The team should develop a **safe and useful product** for the customer rather than hiding problems.

Benefits of Ethical Behavior

- Builds **trust**
- Improves **teamwork**
- Increases **customer satisfaction**
- Reduces conflicts
- Improves **software quality**
- Creates a positive working environment

Easy way to remember:

H-R-T-R-C

H → Honesty
R → Respect
T → Transparency
R → Responsibility
C → Customer interest

Conclusion: Ethics are important in Agile because Agile depends on **trust, transparency, respect, and teamwork**.